

SHETH LUJ COLLEGE

PRACTICAL - 5

Aim:-Implement the Diffie-Hellman key exchange algorithm to securely exchange keys between two entities over an insecure network.

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C:\Python3\Python3\Python3.py [Jupyter]
File Edit Format Run Options Window Help

# Diffie-Hellman Key Exchange Algorithm

p = 23
g = 5

print("Diffie-Hellman Key Exchange")
print("-----")

print("Public Prime (p):", p)
print("Public Base (g):", g)

# Generate a private key
alice_private = int(input("Enter Alice's private key: "))

# Generate a private key
bob_private = int(input("Enter Bob's private key: "))

# Calculate Alice's public key
alice_public = pow(g, alice_private, p)

# Calculate public key
bob_public = pow(g, bob_private, p)

print("Alice's Public Key:", alice_public)
print("Bob's Public Key:", bob_public)

# Calculate shared secret key
alice_shared_secret = pow(bob_public, alice_private, p)
bob_shared_secret = pow(alice_public, bob_private, p)

print("Alice's Shared Secret Key:", alice_shared_secret)
print("Bob's Shared Secret Key:", bob_shared_secret)

# Verify whether both keys are equal
if alice_shared_secret == bob_shared_secret:
    print("Key Exchange Successful!")
    print("Both Alice and Bob have the same secret key.")
else:
    print("Key Exchange Failed!")

```

OUTPUT

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Python 3.10.6 [tags/c3.10.6.final0, Jun 10 2022, 10:26:10] [AMD64 v1.044 64 bit (AMD64)] on win32
Enter "help()" below or click "Help" above for more information.

>>> ===== RESTART: C:/Python3/Python3.py =====
Diffie-Hellman Key Exchange
-----
Public Prime (p): 23
Public Base (g): 5

Enter Alice's private key: 8
Enter Bob's private key: 5

Alice's Public Key: 11
Bob's Public Key: 20

Alice's Shared Secret Key: 3
Bob's Shared Secret Key: 3

Key Exchange Successful!
Both Alice and Bob have the same secret key.
>>>

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